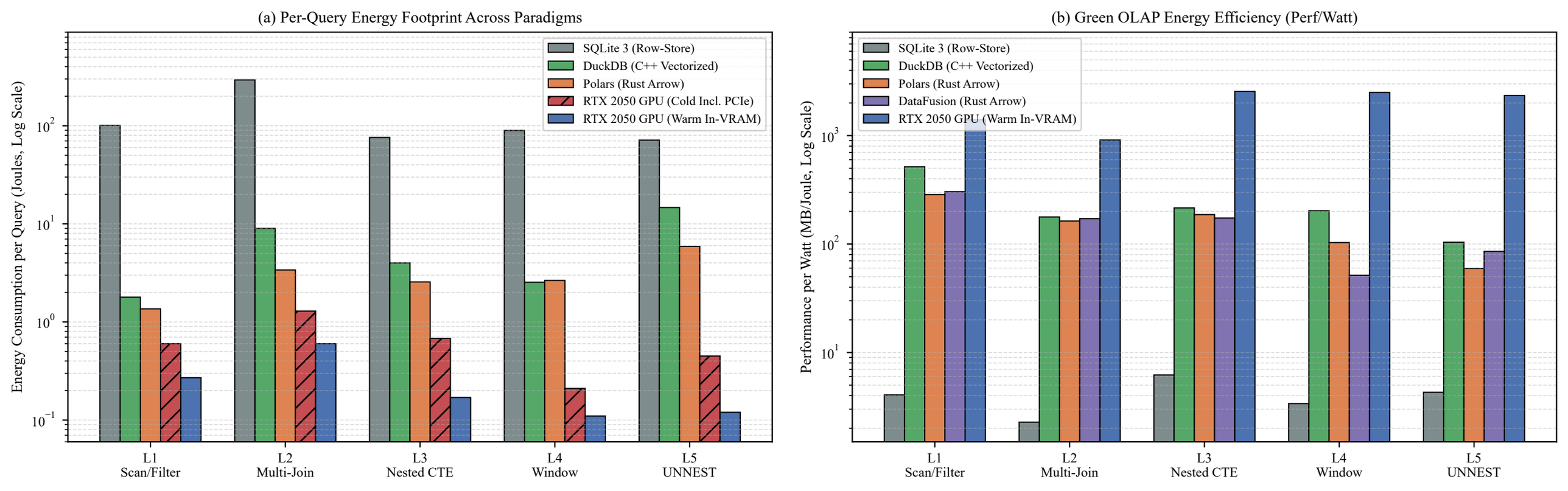
(a) Per-Query Energy Footprint Across Paradigms
Energy Consumption per Query (Joules, Log Scale)
10^{2}
10^{1}
10^{0}
10^{-1}
SQLite 3 (Row-Store)
DuckDB (C++ Vectorized)
Polars (Rust Arrow)
RTX 2050 GPU (Cold Incl. PCIe)
RTX 2050 GPU (Warm In-VRAM)
L1 Scan/Filter
L2 Multi-Join
L3 Nested CTE
L4 Window
L5 UNNEST
(b) Green OLAP Energy Efficiency (Perf/Watt)
Performance per Watt (MB/Joule, Log Scale)
10^{3}
10^{2}
10^{1}
SQLite 3 (Row-Store)
DuckDB (C++ Vectorized)
Polars (Rust Arrow)
DataFusion (Rust Arrow)
RTX 2050 GPU (Warm In-VRAM)
L1 Scan/Filter
L2 Multi-Join
L3 Nested CTE
L4 Window
L5 UNNEST